CSE 4621: Machine Learning
Evaluation Metrics

Lecture 7

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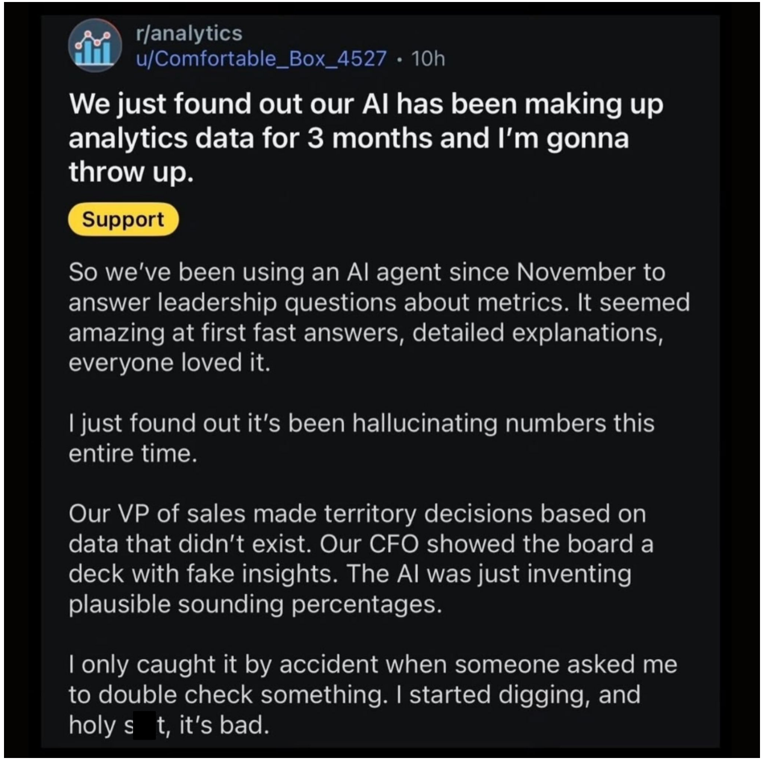
Lecture Outline

- Evaluation Framework
- Regression Metrics
  - MAE, MSE, RMSE
  - R^2, Adjusted R^2, MAPE
- Classification Metrics
  - TP, FP, TN, FN
  - Accuracy, Precision, Recall
  - ROC, AUC

Horror Stories

A cancer screening model classifies every patient as *healthy*. It is trained on a dataset where only 1% have cancer, it achieves 99% accuracy but never catches a single cancer patient

Every metric answers a specific question. You must ask the *right* question for your problem. The metric you optimise during training shapes what your model learns to do.


r/analytics
u/Comfortable_Box_4527 • 10h

We just found out our AI has been making up analytics data for 3 months and I'm gonna throw up.

Support

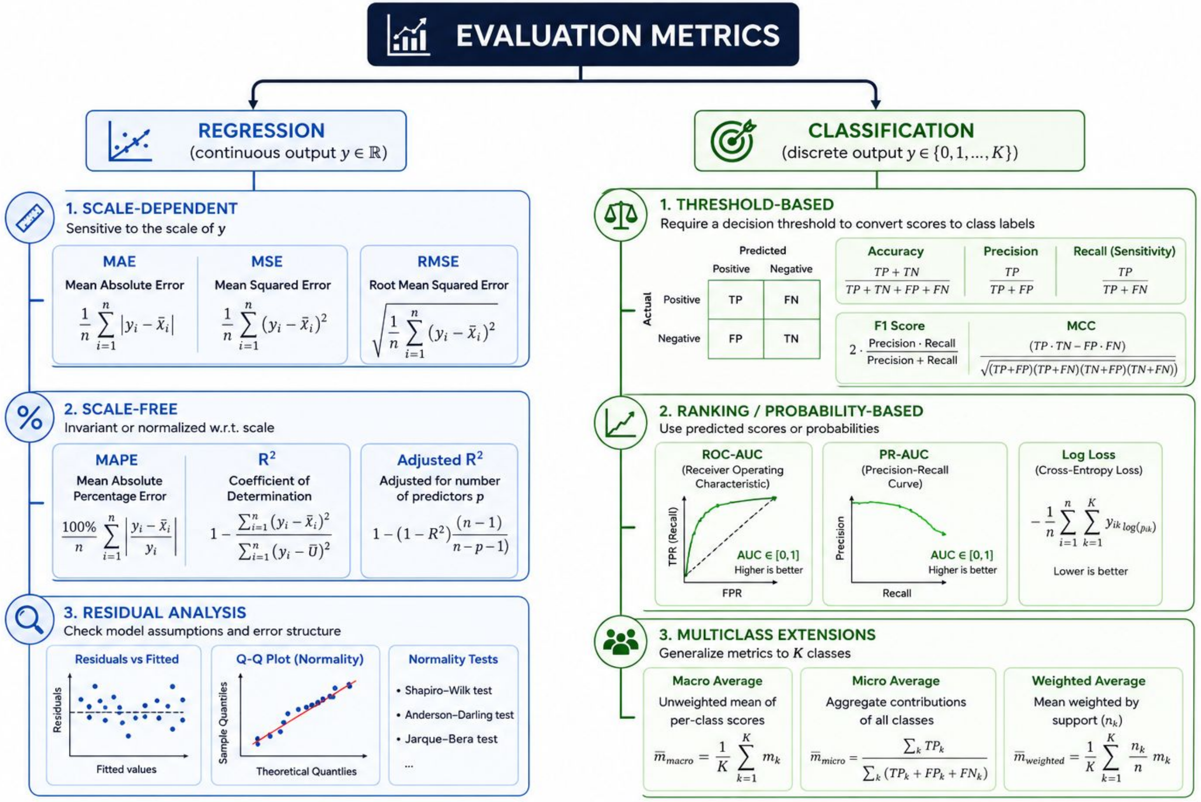
So we've been using an AI agent since November to answer leadership questions about metrics. It seemed amazing at first fast answers, detailed explanations, everyone loved it.

I just found out it's been hallucinating numbers this entire time.

Our VP of sales made territory decisions based on data that didn't exist. Our CFO showed the board a deck with fake insights. The AI was just inventing plausible sounding percentages.

I only caught it by accident when someone asked me to double check something. I started digging, and holy s[redacted]t, it's bad.

Evaluation Framework



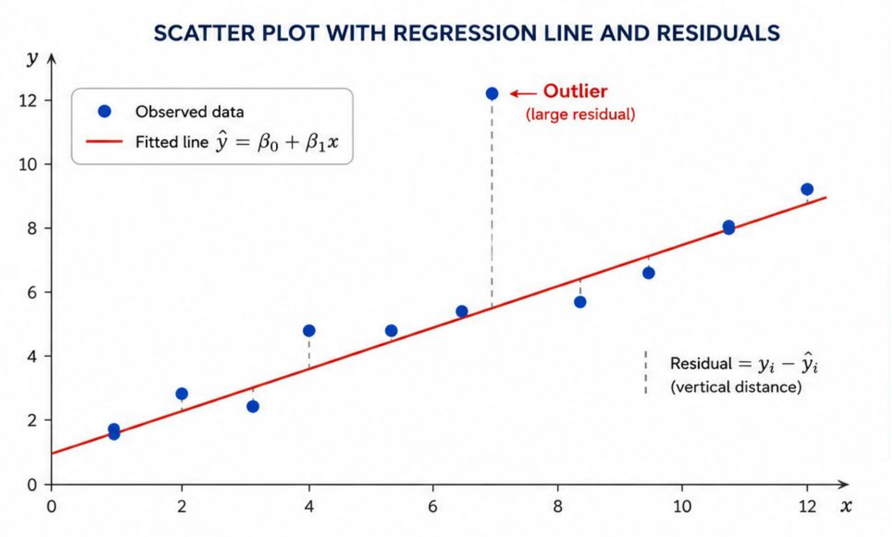
Regression Metrics

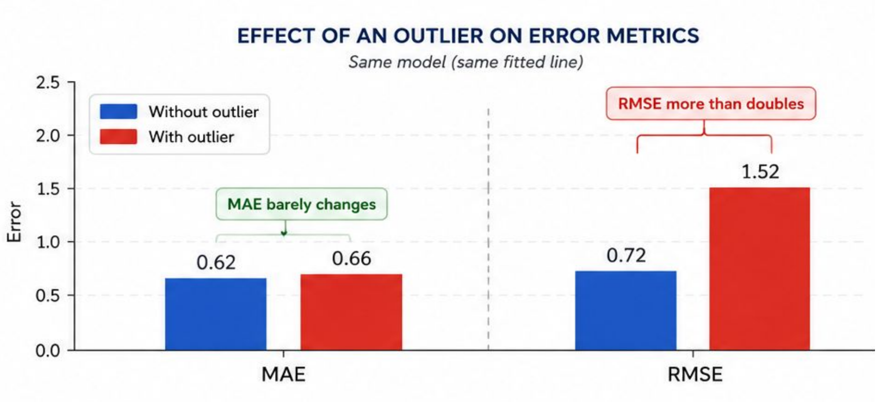
- Mean Absolute Error (MAE): $\mathrm{MAE} = \frac{1}{m}\sum_{i=1}^{m} |y^{(i)} - \hat{y}^{(i)}|$

- Mean Squared Error (MSE): $\mathrm{MSE} = \frac{1}{m}\sum_{i=1}^{m} (y^{(i)} - \hat{y}^{(i)})^2$

- Root Mean Squared Error (RMSE): $\mathrm{RMSE} = \sqrt{\frac{1}{m}\sum_{i=1}^{m} (y^{(i)} - \hat{y}^{(i)})^2}$

Regression Metrics





Regression Metrics

How much of the variation in the target variable is explained by the model?

R^2 (Coefficient of Determination):

- If every house sold for exactly the same price, there would be no variation to explain
- R^2 tells you what fraction of that variation your model captures

$$R^2 = 1 - \frac{\sum_{i=1}^{m}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{m}(y_i - \bar{y}_i)^2}$$

R^2	Meaning
1.0	Perfect predictions
0.8	Model explains 80% of variance
0.0	No better than predicting the mean
< 0	Worse than predicting the mean

Regression Metrics

Adjusted R^2:

- R^2 always increases or stays the same when you add more features, even if those features are useless
- Adjusted R^2 penalizes models for adding unnecessary variables

$$\bar{R}^2 = 1 - (1 - R^2)\frac{m-1}{m-n-1}$$

Regression Metrics

Mean Absolute Percentage Error (MAPE):

- Measures average prediction error as a percentage
- It's easy to explain but can break when percentage error becomes undefined because you'd divide by zero

MAPE	Quality
$< 5\%$	Excellent
5–10%	Very good
10–20%	Reasonable
$> 20\%$	Often poor

$$\text{MAPE} = \frac{100}{m} \sum_{i=1}^{m} \left| \frac{y_i - \hat{y}_i}{y_i} \right|$$

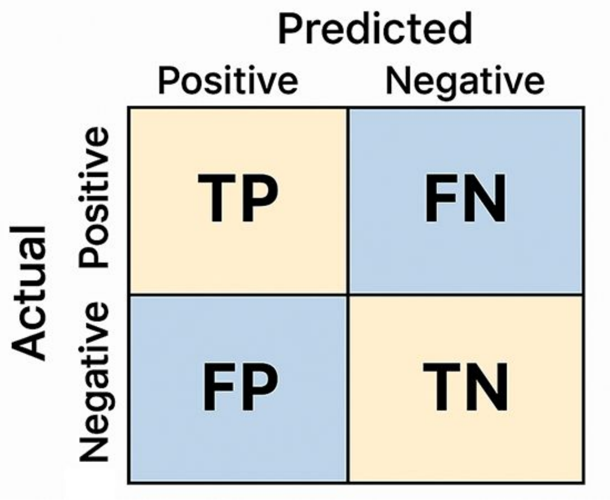
Classification Metrics

- True Positive (TP): Predicted positive, actually positive
- False Positive (FP / Type I Error): Predicted positive, actually negative
- False Negative (FN / Type II Error): Predicted negative, actually positive
- True Negative (TN): Predicted negative, actually negative

In total, there are 100 patients, where 10 have disease. Model predicts 8 of the sick correctly flagged, 2 missed, 15 healthy people incorrectly flagged, 75 correctly cleared

Classification Metrics

- True Positive (TP): Predicted positive, actually positive
- False Positive (FP / Type I Error): Predicted positive, actually negative
- False Negative (FN / Type II Error): Predicted negative, actually positive
- True Negative (TN): Predicted negative, actually negative



Domain	FP cost	FN cost	Which is worse?
Spam filter	Lose important email	See some spam	FP worse
Cancer screening	Unnecessary biopsy	Missed cancer	FN far worse
Fraud detection	Declined valid transaction	Fraud goes through	Depends on amount
Self-driving car	Unnecessary brake	Missed pedestrian	FN catastrophically worse

Classification Metrics

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \qquad \text{Precision} = \frac{TP}{TP + FP} \qquad \text{Recall} = \frac{TP}{TP + FN}$$

$$\text{F1 Score} = \frac{2PR}{P + R} = \frac{2TP}{2TP + FP + FN}$$

In total, there are 100 patients, where 10 have disease. Model predicts 8 of the sick correctly flagged (TP=8), 2 missed (FN=2), 15 healthy people incorrectly flagged (FP=15), 75 correctly cleared (TN=75)

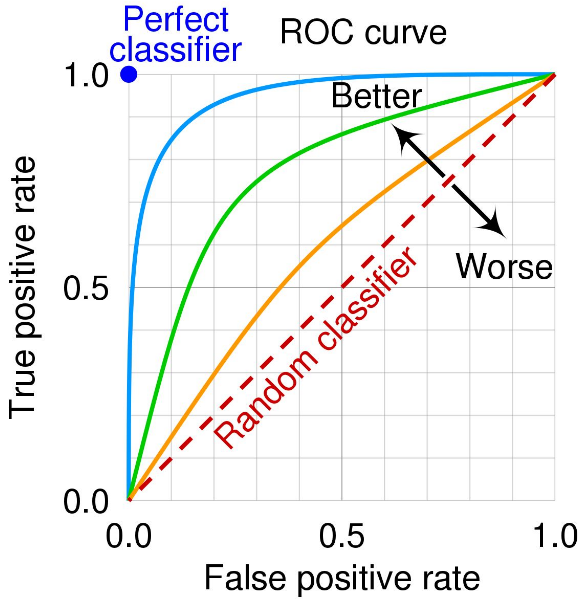
Classification Metrics

ROC Curve (Receiver Operating Characteristic)

- How well a classifier separates the positive class from the negative class across different decision thresholds

In the X-axis → False positive rate $\mathrm{FPR} = \frac{FP}{FP + TN}$

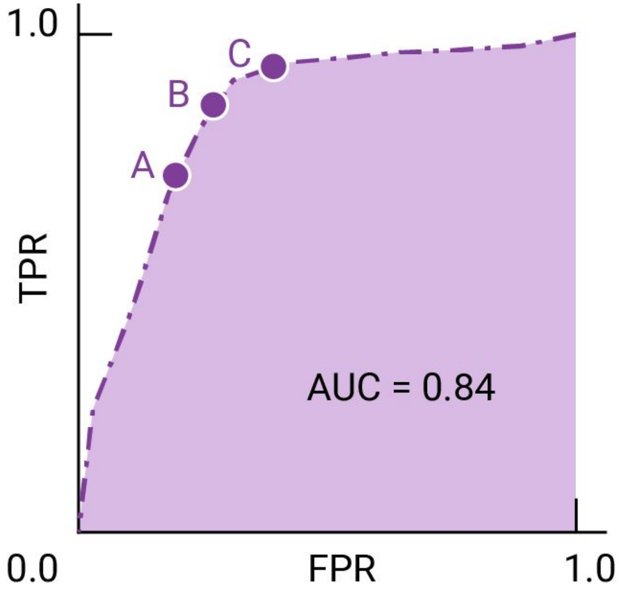
In the Y-axis → True positive rate (Recall) $\mathrm{TPR} = \frac{TP}{TP + FN}$



Classification Metrics

AUC (Area Under the ROC Curve)

- AUC summarizes the entire ROC curve into a single number
- The probability that the model ranks a randomly chosen positive example higher than a randomly chosen negative example



Classification Metrics

When one class dominates (e.g., 95% negative), simple accuracy is misleading. A model predicting "always negative" gets 95% accuracy without learning anything

$$\text{Balanced Accuracy} = \frac{TPR + TNR}{2}$$

Average of recall across both classes; more appropriate than accuracy when classes are unbalanced

- Oversample minority: SMOTE, RandomOverSampler
- Undersample majority
- Use class weights in loss

Textbook Chapters

- [Goodfellow] - Chapter 5.3, 5.5
- [Bishop] - Chapter 4.1
- [Alpaydin] - Chapter 19

Thank you